Right Upper Quadrant Pain

EVIDENCE TABLE

	Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
1.	Hindman NM, Arif-Tiwari H, Kamel IR, et al. ACR Appropriateness Criteria® Jaundice. J Am Coll Radiol 2019;16:S126-S40.	Review/Other-Dx	N/A	Evidence-based guidelines to assist referring physicians and other providers in making the most appropriate imaging or treatment decision for jaundice.	No results stated in abstract.	4
2.	Scheirey CD, Fowler KJ, Therrien JA, et al. ACR Appropriateness Criteria® Acute Nonlocalized Abdominal Pain. J Am Coll Radiol 2018;15:S217-S31.	Review/Other-Dx	N/A	Evidence-based guidelines to assist referring physicians and other providers in making the most appropriate imaging or treatment decision for acute nonlocalized abdominal pain.	No results stated in abstract.	4
3.	Vij A, Zaheer A, Kamel IR, et al. ACR Appropriateness Criteria® Epigastric Pain. J Am Coll Radiol 2021;18:S330-S39.	Review/Other-Dx	N/A	Evidence-based guidelines to assist referring physicians and other providers in making the most appropriate imaging or treatment decision for epigastric pain.	No results stated in abstract.	4
4.	Porter KK, Zaheer A, Kamel IR, et al. ACR Appropriateness Criteria® Acute Pancreatitis. J Am Coll Radiol 2019;16:S316-S30.	Review/Other-Dx	N/A	Evidence-based guidelines to assist referring physicians and other providers in making the most appropriate imaging or treatment decision for acute pancreatitis.	No results stated in abstract.	4
5.	Chang KJ, Marin D, Kim DH, et al. ACR Appropriateness Criteria® Suspected Small-Bowel Obstruction. J Am Coll Radiol 2020;17:S305-S14.	Review/Other-Dx	N/A	Evidence-based guidelines to assist referring physicians and other providers in making the most appropriate imaging or treatment decision for suspected small-bowel obstruction.	No results stated in abstract.	4
6.	Wadhwa V, Jobanputra Y, Garg SK, Patwardhan S, Mehta D, Sanaka MR. Nationwide trends of hospital admissions for acute cholecystitis in the United States. Gastroenterol Rep (Oxf) 2017;5:36-42.	Review/Other-Dx	149661 patients	To use a national database of US hospitals to evaluate the incidence and costs of hospital admissions associated with acute cholecystitis.	In 1997, there were 149 661 hospital admissions with a principal discharge diagnosis of acute cholecystitis, which increased to 215 995 in 2012 ($P < 0.001$). The mean length of stay for acute cholecystitis decreased by 17% between 1997 and 2012 (i.e. from 4.7 days to 3.9 days; ($P < 0.05$). During the same time period, however, mean hospital charges have increased by 195.4 % from US$14 608 per patient in 1997 to US$43 152 per patient in 2012 ($P < 0.001$).	4
7.	Bennett GL. Evaluating Patients with Right Upper Quadrant Pain. Radiol Clin North Am. 2015;53(6):1093-1130.	Review/Other-Dx	N/A	To discuss the complimentary roles of various imaging modalities in the evaluation of the patient with right upper quadrant pain.	No results stated in abstract.	4

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8. Trowbridge RL, Rutkowski NK, Shojania KG. Does this patient have acute cholecystitis? JAMA. 2003; 289(1):80-86.	Review/Other-Dx	17 studies	To determine role of clinical or laboratory testing in identifying patients who require diagnostic imaging tests to rule in or rule out the diagnosis of AC.	No clinical criteria had high or low likelihood ratio for AC except Murphy sign (positive likelihood ratio 2.8) and right upper quadrant tenderness (positive likelihood ratio 0.4).	4
9. Wertz JR, Lopez JM, Olson D, Thompson WM. Comparing the Diagnostic Accuracy of Ultrasound and CT in Evaluating Acute Cholecystitis. AJR Am J Roentgenol. 211(2):W92-W97, 2018 08.	Observational-Dx	60 patients	To evaluate the diagnostic accuracies of ultrasound (US) and CT.	The sensitivity of CT for detecting AC was significantly greater than that of US: 85% versus 68% (p = 0.043), respectively; however, the negative predictive values of CT and US did not differ significantly: 90% versus 77% (p = 0.24-0.26). Because there were no false-positives, the specificity and positive predictive values for both modalities were 100%. Among the 42 patients who underwent CT and US, both modalities were positive for AC in 25 patients, CT was positive and US was negative in 10 patients, and US was positive and CT was negative in two patients; in five patients, both US and CT were negative.	3
10. Hwang SH, You JS, Song MK, Choi JY, Kim MJ, Chung YE. Comparison of diagnostic performance between single- and multiphasic contrast-enhanced abdominopelvic computed tomography in patients admitted to the emergency department with abdominal pain: potential radiation dose reduction. European Radiology. 25(4):1048-58, 2015 Apr.	Observational-Dx	253 patients	To evaluate feasibility of radiation dose reduction by optimal phase selection of computed tomography (CT) in patients who visited the emergency department (ED) for abdominal pain.	There was no difference in diagnostic performance among three image sets, although diagnostic confidence level was significantly improved after review of triphasic images compared with both HVP images only or HVP with precontrast images (confidence scale, 4.64 +/- 0.05, 4.66 +/- 0.05, and 4.76 +/- 0.04 in the order of the sets; overall P = 0.0008). Similar trends were observed in the subgroup analysis for diagnosis of pelvic inflammatory disease and cholecystitis.	3

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11. Uyeda JW, Trinh TW, Wortman JR, Sodickson AD. The Dual Energy Hot Gallbladder and Rim Signs: Evaluation of DECT Iodine Content in Acute Cholecystitis. Paper presented at: Radiological Society of North America Scientific Assembly and Annual Meeting Program., 2016; Oak Brook, Ill. Available at: http://archive.rsna.org/2016/16006088.html.	Review/Other-Dx	17 patients	To assess differences in the dual energy CT (DECT) iodine content of pericholecystic hepatic parenchyma and of the gallbladder wall in acute cholecystitis compared with controls.	There was no significant difference between cholecystitis and control patients in normalized iodine content within the hepatic parenchyma remote from the gallbladder fossa (p=0.72). However, compared with controls, acute cholecystitis patients demonstrated higher normalized iodine concentration values within the hepatic parenchyma of the gallbladder fossa and within the gallbladder wall (p<0.001 for both comparisons).	4
12. Hallal AH, Amortegui JD, Jeroukhimov IM, et al. Magnetic resonance cholangiopancreatography accurately detects common bile duct stones in resolving gallstone pancreatitis. J Am Coll Surg 2005;200:869-75.	Observational-Dx	63 patients	To evaluate magnetic resonance cholangiopancreatography (MRCP) in detecting choledocholithiasis in this group of patients.	Sixty-three patients were randomized (34 to group 1 and 29 to group 2). CBD stones were found in 5 patients in group 1. CBD exploration was performed in 2 patients, preoperative ERCP in 1, and postoperative ERCP in the other 2. MRCP showed CBD stones in 4 patients in group 2. There were two false-positive MRCPs. Four patients with a negative MRCP did not have IOC or ERCP, the remaining 21 patients with a negative MRCP had a negative IOC. The MRCP sensitivity was 100% (95% CI, 16-100%), specificity 91% (95% CI, 72-99%), positive predictive value 50% (95% CI, 7-93%), negative predictive value 100% (95% CI, 84-100%), and accuracy 92% (95% CI, 74-99%).	3
13. Tenner S, Dubner H, Steinberg W. Predicting gallstone pancreatitis with laboratory parameters: a meta-analysis. Am J Gastroenterol 1994;89:1863-6.	Meta-analysis	N/A	To define the usefulness of biochemical laboratory values in distinguishing gallstone from non-gallstone acute pancreatitis	Using receiver operating characteristic curves for each of these four parameters, we determined that the ALT level was the most clinically useful parameter. The higher the serum level of ALT, the greater its specificity and positive predictive value in diagnosing gallstone pancreatitis. At ALT levels greater than or equal to 150 IU/L (approximately a 3-fold elevation), the probability of gallstone pancreatitis is 95%.	Inadequate

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14. Alobaidi M, Gupta R, Jafri SZ, Fink-Bennet DM. Current trends in imaging evaluation of acute cholecystitis. Emerg Radiol. 2004; 10(5):256-258.	Observational-Dx	117 patients	To compare roles of US to HIDA in the diagnosis of AC.	Patients had combinations of US, HIDA, both or no imaging. The best results were HIDA imaging with sensitivity of 91% and US has sensitivity of 62%.	2
15. Bennett GL, Balthazar EJ. Ultrasound and CT evaluation of emergent gallbladder pathology. Radiol Clin North Am. 2003; 41(6):1203-1216.	Review/Other-Dx	N/A	Review US and CT evaluation of emergent gallbladder pathology.	US is the initial imaging modality of choice for the evaluation of suspected acute gallbladder disorders. CT also plays an important role in the evaluation of acute gallbladder pathology and is useful where US findings are equivocal. CT is also extremely valuable in the assessment of suspected complications of AC, particularly emphysematous cholecystitis, hemorrhagic cholecystitis, and gallbladder perforation, which are often very difficult diagnoses to establish at US.	4
16. Hanbidge AE, Buckler PM, O'Malley ME, Wilson SR. From the RSNA refresher courses: imaging evaluation for acute pain in the right upper quadrant. Radiographics. 2004; 24(4):1117-1135.	Review/Other-Dx	N/A	To review imaging features of AC.	US is the main imaging modality for assessment of acute right upper quadrant pain. US is sensitive and specific in demonstrating gallstones, biliary dilatation, and features that suggest acute inflammatory disease.	4
17. Smith EA, Dillman JR, Elsayes KM, Menias CO, Bude RO. Cross-sectional imaging of acute and chronic gallbladder inflammatory disease. AJR. 2009; 192(1):188-196.	Review/Other-Dx	N/A	Comprehensive review of the clinical and cross-sectional imaging features of different acute and chronic gallbladder inflammatory diseases.	Inflammatory gallbladder diseases are a common source of abdominal pain and cause considerable morbidity and mortality. Numerous other gallbladder inflammatory conditions may also occur that can be readily diagnosed by cross-sectional imaging.	4

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
18. van Randen A, Lameris W, Luitse JS, et al. The role of plain radiographs in patients with acute abdominal pain at the ED. Am J Emerg Med. 2011;29(6):582-589 e582.	Observational-Dx	1,021 patients	Prospective multicenter trial to evaluate the added value of plain radiographs on top of clinical assessment in unselected patients presenting with acute abdominal pain at the emergency department.	In 117 of 1,021 patients, the diagnosis changed after plain radiographs, and this change was correct in 39 patients (22% of changed diagnoses and 4% of total study population). Overall, the clinical diagnosis was correct in 502 (49%) patients. The diagnosis after evaluation of the radiographs was correct in 514 (50%) patients, a nonsignificant difference (P=.14). In 65% of patients with unchanged diagnosis before and after plain radiography, the level of confidence of that diagnosis did not change either. The added value of plain radiographs is too limited to advocate their routine use in the diagnostic workup of patients with acute abdominal pain, because few diagnoses change and the level of confidence were mostly not affected.	3
19. Ahn SH, Mayo-Smith WW, Murphy BL, Reinert SE, Cronan JJ. Acute nontraumatic abdominal pain in adult patients: abdominal radiography compared with CT evaluation. Radiology. 2002 Oct;225(1):159-64.	Observational-Dx	871 patients had abdominal radiography, and 188 patients had abdominal CT	Retrospective study to compare the diagnostic yield of abdominal radiography with that of CT in adult patients presenting to the emergency department with nontraumatic abdominal pain.	Interpretation of abdominal radiographs was nonspecific in 588 (68%) of 871 patients, normal in 200 (23%), and abnormal in 83 (10%). The highest sensitivity of abdominal radiography was 90% for intra-abdominal foreign body and 49% for bowel obstruction. Abdominal radiography had 0% sensitivity for appendicitis, pyelonephritis, pancreatitis, and diverticulitis. Sensitivities of abdominal CT were highest for bowel obstruction and urolithiasis at 75% and 68%, respectively. Abdominal radiographs are not sensitive in the evaluation of adult patients presenting to the emergency department with nontraumatic abdominal pain.	3

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
20. Prasannan S, Zhueng TJ, Gul YA. Diagnostic value of plain abdominal radiographs in patients with acute abdominal pain. Asian J Surg 2005;28:246-51.	Observational-Dx	168 patients	To evaluate the appropriateness and contribution of plain abdominal radiographs (PAR) in the diagnosis and management of adult patients presenting with acute abdominal pain.	Of 168 patients (246 PAR), 59 (35%) had positive findings on PAR. PAR were most sensitive in cases of intestinal obstruction (odds ratio, OR = 33.548, r = 0.561). The sensitivity was further increased if three of the following predictive signs were present: exaggerated bowel sounds (OR = 13.350, r = 0.154), abdominal distension (OR = 2.993, r = 0.234) and age over 50 years (OR = 2.301, r = 0.027). PAR were non-diagnostic in 82% of patients with acute abdominal pain ($p < 0.001$).	3
21. Zeina AR, Shapira-Rootman M, Mahamid A, Ashkar J, Abu-Mouch S, Nachtigal A. Role of Plain Abdominal Radiographs in the Evaluation of Patients with Non-Traumatic Abdominal Pain. Isr Med Assoc J 2015;17:678-81.	Observational-Dx	573 patients	To describe the frequency and outcomes of the use of plain abdominal radiographs in the diagnosis of patients presenting with acute non-traumatic abdominal pain in the ED of a medical center.	Of 573 consecutive patients, 300 (52%) underwent abdominal radiography. Findings were normal in 88% (n = 264), non-specific in 7.3% (n = 22), and abnormal in 4.7% (n = 14). For those with normal results, no further imaging was ordered for 43% (114/264). Of the 57% (150/264) who had follow-up imaging, 65% (98/150) showed abnormal findings. In 9 (3%) of the 300 patients, abdominal radiography identified bowel perforations and obstructions, and treatment was provided without the need for further radiologic examination.	4
22. Boland GW, Slater G, Lu DS, Eisenberg P, Lee MJ, Mueller PR. Prevalence and significance of gallbladder abnormalities seen on sonography in intensive care unit patients. AJR. 2000; 174(4):973-977.	Observational-Dx	55 patients	To evaluate sonographic abnormalities of the gallbladder other than acalculous cholecystitis across a broad range of intensive care unit patients.	11/55 patients were found to have gallbladder calculi and were excluded from the study. 37 (84%) of the remaining 44 patients had at least one sonographic abnormality while in the intensive care unit. 25 (57%) of the 44 patients had as many as three abnormalities found on sonography, and 6 (14%) of 44 patients had 4 or 5 sonographic findings of gallbladder abnormalities while in the intensive care unit. No statistically significant correlation was found among any of these sonographic abnormalities and the clinical and laboratory parameters.	2

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23. Lameris W, van Randen A, van Es HW, et al. Imaging strategies for detection of urgent conditions in patients with acute abdominal pain: diagnostic accuracy study. BMJ. 2009;338:b2431.	Observational-Dx	1,021 patients	Multicentre diagnostic accuracy study with prospective data collection to identify an optimal imaging strategy for the accurate detection of urgent conditions in patients with acute abdominal pain.	661 (65%) patients had a final diagnosis classified as urgent. The initial clinical diagnosis resulted in many false positive urgent diagnoses, which were significantly reduced after US or CT. CT detected more urgent diagnoses than did US: sensitivity was 89% (95% CI, 87% to 92%) for CT and 70% (67% to 74%) for US (P<0.001). A conditional strategy with CT only after negative or inconclusive US yielded the highest sensitivity, missing only 6% of urgent cases. With this strategy, only 49% (46% to 52%) of patients would have CT. Alternative strategies guided by body mass index, age, or location of the pain would all result in a loss of sensitivity. Although CT is the most sensitive imaging investigation for detecting urgent conditions in patients with abdominal pain, using US first and CT only in those with negative or inconclusive US results in the best sensitivity and lowers exposure to radiation.	3
24. MacKersie AB, Lane MJ, Gerhardt RT, et al. Nontraumatic acute abdominal pain: unenhanced helical CT compared with three-view acute abdominal series. Radiology. 2005;237(1):114-122.	Observational-Dx	91 patients	Prospective, blinded study to compare the diagnostic accuracy of unenhanced CT with a three-view acute abdominal series in patients with acute non-traumatic abdominal pain.	CT yielded sensitivity, specificity, and accuracy of 96.0%, 95.1%, and 95.6%, respectively. Acute abdominal series interpretations yielded sensitivity, specificity, and accuracy of 30.0%, 87.8%, and 56.0%, respectively. CT is the imaging procedure of choice.	1

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25. Kiewiet JJ, Leeuwenburgh MM, Bipat S, Bossuyt PM, Stoker J, Boermeester MA. A systematic review and meta-analysis of diagnostic performance of imaging in acute cholecystitis. Radiology. 2012; 264(3):708-720.	Meta-analysis	57 studies, 5859 patients	To update previously summarized estimates of diagnostic accuracy for AC and to obtain summary estimates for more recently introduced modalities.	Sensitivity of cholescintigraphy (96%; 95% CI: 94%, 97%) was significantly higher than sensitivity of US (81%; 95% CI: 75%, 87%) and MRI (85%; 95% CI: 66%, 95%). There were no significant differences in specificity among cholescintigraphy (90%; 95% CI: 86%, 93%), US (83%; 95% CI: 74%, 89%) and MRI (81%; 95% CI: 69%, 90%). Only one study about evaluation of CT met the inclusion criteria; the reported sensitivity was 94% (95% CI: 73%, 99%) at a specificity of 59% (95% CI: 42%, 74%).	Good
26. Marasco G, Verardi FM, Eusebi LH, et al. Diagnostic imaging for acute abdominal pain in an Emergency Department in Italy. Intern. emerg. medicine. 14(7):1147-1153, 2019 10.	Observational-Dx	578 patients	To evaluate the use and diagnostic performance of imaging techniques in adult patients with AAP in an ED in Italy.	Of the 578 patients (female 52.8%, mean age 51.3 years) admitted to the ED for AAP, 433 (74.9%) underwent abdominal imaging. The most frequent techniques used were abdominal plain radiography and ultrasonography (US), performed in 38.4% and 37.9% of patients, respectively, followed by computed tomography (CT) in 28% of patients. Plain radiography yielded a sensitivity of 28% and specificity of 91.1%; the sensitivity increased to 79.4% in patients with small bowel obstruction. Ultrasonography's sensitivity and specificity were 61.8% and 98.4%, respectively; the sensitivity of US reached 85.2% and 90% in patients with acute cholecystitis/biliary colic and urolithiasis, respectively. The sensitivity and specificity of CT were 87.8% and 92.9%, respectively.	3
27. Revzin MV, Scoutt LM, Garner JG, Moore CL. Right Upper Quadrant Pain: Ultrasound First! J Ultrasound Med 2017;36:1975-85.	Review/Other-Dx	N/A	To discuss the importance of ultrasound in right upper quadrant pain.	No results stated in abstract.	4

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28. Liu TH, Consorti ET, Kawashima A, et al. Patient evaluation and management with selective use of magnetic resonance cholangiography and endoscopic retrograde cholangiopancreatography before laparoscopic cholecystectomy. Ann Surg. 2001;234(1):33-40.	Observational-Dx	440 patients	To assess the utility of triage guidelines for patients with cholelithiasis and suspected choledocholithiasis, incorporating selective use of magnetic resonance cholangiography (MRC) and endoscopic retrograde cholangiopancreatography (ERCP) before laparoscopic cholecystectomy (LC).	Choledocholithiasis was detected in 43 of 440 patients (9.8%). The occurrence of choledocholithiasis among patients in the four groups were 92.6% (25/27), 32.4% (12/37), 3.8% (2/52), and 0.9% (3/324) for groups 1, 2, 3, and 4, respectively (P <.001). MRC was used for 8.4% (37/440) of patients. Patient triage resulted in the identification of common bile duct stones during preoperative ERCP in 92.3% (36/39) of the patients. Unsuspected common bile duct stones occurred in six patients (1.4%).	3
29. Grand D, Horton KM, Fishman EK. CT of the gallbladder: spectrum of disease. AJR Am J Roentgenol 2004;183:163-70.	Review/Other-Dx	N/A	To study CT of the gall bladder.	No results stated in abstract.	4
30. Patel NB, Oto A, Thomas S. Multidetector CT of emergent biliary pathologic conditions. Radiographics. 2013;33(7):1867-1888.	Review/Other-Dx	N/A	To review CT techniques useful for imaging the biliary tract in patients with acute abdominal pain and to describe clinical signs and symptoms; typical imaging findings; and interventional radiologic, endoscopic, and surgical treatment of common and uncommon acute biliary pathologic conditions.	No results stated in abstract.	4
31. Bennett GL, Rusinek H, Lisi V, et al. CT findings in acute gangrenous cholecystitis. AJR. 2002; 178(2):275-281.	Observational-Dx	75 patients; 4 observers	Retrospective review of CT scans to determine value of CT for gangrenous cholecystitis.	Best criteria for gangrenous cholecystitis were air in wall or lumen, intraluminal membranes, irregular or absent wall, and abscess. Absence of wall enhancement, pericholecystic fluid and gall bladder distention. Overall accuracy of CT, 87%.	3

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32. De Vargas Macciucca M, Lanciotti S, De Cicco ML, Coniglio M, Gualdi GF. Ultrasonographic and spiral CT evaluation of simple and complicated acute cholecystitis: diagnostic protocol assessment based on personal experience and review of the literature. Radiol Med. 2006; 111(2):167-180.	Observational-Dx	35 patients	Retrospective review to identify and classify the US and CT signs of simple and complicated AC and to define the correct diagnostic protocol. Gold standard used was histological findings.	US had accuracy of 66.6%, sensitivity of 37.5%, specificity of 70%, PPV of 100%, and an NPV of 58.3%. CT had 100% accuracy, sensitivity, and specificity. If more than two major signs associated with one minor sign or at least one sign of complication are present at US, CT is mandatory to recognize and evaluate the type of complication and indicate appropriate treatment.	3
33. Kim YK, Kwak HS, Kim CS, et al. CT findings of mild forms or early manifestations of acute cholecystitis. Clin Imaging. 2009; 33(4):274-280.	Observational-Dx	34 patients with mild or early AC and 34 control	To determine the most predictive CT feature of the mild forms or early manifestations of AC. Two radiologists analyzed CT of patients.	Most significant predictor of mild or early AC on CT was the presence of pericholecystic increased attenuation on the arterial phase (sensitivity, 82.4%), followed by indistinctness of the interface between the gall bladder and liver (sensitivity, 38.0%), which were identified by both observers with good agreement.	2
34. Shakespear JS, Shaaban AM, Rezvani M. CT findings of acute cholecystitis and its complications. AJR. 2010; 194(6):1523-1529.	Review/Other-Dx	N/A	To describe and illustrate the CT findings of AC and its complications.	CT findings suggesting AC should be interpreted with caution and should probably serve as justification for further investigation with abdominal US. CT has a relatively high NPV, and AC is unlikely in the setting of a negative CT. Complications of AC have a characteristic CT appearance and include necrosis, perforation, abscess formation, intraluminal hemorrhage, and wall emphysema.	4

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35. Tsai MJ, Chen JD, Tiu CM, Chou YH, Hu SC, Chang CY. Can acute cholecystitis with gallbladder perforation be detected preoperatively by computed tomography in ED? Correlation with clinical data and computed tomography features. Am J Emerg Med. 2009; 27(5):574-581.	Observational-Dx	75 patients	Retrospective review to determine which CT findings and clinical data can help diagnose gallbladder perforation in AC. Medical records and CT findings were compared between 2 groups with and without gallbladder perforation.	16 patients had gallbladder perforation. Higher mortality rate was found in the perforation group (18.8% vs 1.7%; P=.029). Older age (>70 years; P=.004) and higher percentage of segmented neutrophil (>80%; P=.027) were significant clinical factors for predicting gallbladder perforation in AC. In multivariate analysis, visualized gallbladder wall defect was the most significant predicting CT feature for diagnosing gallbladder perforation in AC.	3
36. Akpinar E, Turkbey B, Karcaaltincaba M, et al. Initial experience on utility of gadobenate dimeglumine (Gd-BOPTA) enhanced T1-weighted MR cholangiography in diagnosis of acute cholecystitis. J Magn Reson Imaging. 2009; 30(3):578-585.	Review/Other-Dx	11 consecutive patients with acute right upper quadrant pain; 15 controls	Prospective study to examine the feasibility of the use of Gd-BOPTA-enhanced T1-weighted MR cholangiography in diagnosis of AC.	In the control group, Gd-BOPTA was visualized within the gallbladder in all subjects. For the study group, gallstones were present in 9 patients on MR cholangiography. In addition to anatomical assessment, Gd-BOPTA-enhanced MR cholangiography can provide functional evaluation similar to scintigraphy in diagnosing AC in patients with acute right upper quadrant pain and equivocal findings.	4

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37. Altun E, Semelka RC, Elias J, Jr., et al. Acute cholecystitis: MR findings and differentiation from chronic cholecystitis. Radiology. 2007; 244(1):174-183.	Observational-Dx	32 patients; 4 blinded reviewers	To retrospectively determine the sensitivity and specificity of MRI for differentiation between acute and chronic cholecystitis, with histopathologic analysis as the reference standard.	MRI sensitivity and specificity were 95% (18/19 patients) and 69% (9/13 patients), respectively. Sensitivities of increased gallbladder wall enhancement and increased transient pericholecystic hepatic enhancement were 74% (14/19 patients) and 62% (10/16 patients), respectively. Both findings had 92% (12/13 patients) specificity. Sensitivities of increased wall thickness, pericholecystic fluid, and adjacent fat signal intensity changes were 100% (19/19 patients), 95% (18/19 patients), and 95% (18/19 patients), respectively; specificities were 54% (7/13 patients), 38% (5/13 patients), and 54% (7/13 patients), respectively. Pericholecystic abscess, intraluminal membranes, and wall irregularity or defect each had 100% (13/13 patients) specificity; sensitivities were 11% (2/19 patients), 26% (five of 19 patients), and 21% (4/19 patients), respectively. Increased gallbladder wall enhancement (P<.001) and increased transient pericholecystic hepatic enhancement (P=.003) were the most significantly different between acute and chronic cholecystitis.	3
38. Oh KY, Gilfeather M, Kennedy A, et al. Limited abdominal MRI in the evaluation of acute right upper quadrant pain. Abdom Imaging. 2003; 28(5):643-651.	Observational-Dx	24 patients	To investigate whether limited abdominal MRI is as effective as transabdominal US in evaluating patients presenting with acute right upper quadrant pain.	MRI and US demonstrated no statistically significant difference in the diagnosis of gallbladder wall thickening, the presence of gallstones or pericholecystic fluid, or the diagnosis of AC (P>0.05). The sensitivity of both for AC was 50%, with specificities of 89% and 86% for US and MRI, respectively. US readers more frequently requested additional tests and displayed more variability in whether they could adequately see the common bile duct.	2

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39. Cho JY, Han HS, Yoon YS, Ahn KS, Lee SH, Hwang JH. Hepatobiliary scan for assessing disease severity in patients with cholelithiasis. Arch Surg. 2011; 146(2):169-174.	Observational-Dx	941 patients	To evaluate the role of a hepatobiliary scan for predicting the severity of cholecystitis and the difficulty of laparoscopic cholecystectomy.	The overall predictive value of the gallbladder ejection fraction for predicting AC was 82.9% (P<.001), and the sensitivity and specificity of the gallbladder ejection fraction at a set point of 30.0% were 92.1% and 61.6%, respectively. The mean severity of the cholecystitis score and the difficulty in performing laparoscopic cholecystectomy scores in the patients with gallbladder nonvisualization or a gallbladder ejection fraction less than 30.0% (2.9 [2.5] and 0.5 [0.9], respectively) were significantly higher than those for the patients with a gallbladder ejection fraction of 30.0% or higher (0.5 [1.1] and 0.3 [0.6]; P<.001 and P=.01, respectively). Moreover, the patients with gallbladder nonvisualization or a gallbladder ejection fraction >30.0% experienced higher rates of complication after laparoscopic cholecystectomy than did the patients with a gallbladder ejection fraction of 30.0% or higher (6.3% vs 2.6%; P=.006).	3
40. Laing FC, Federle MP, Jeffrey RB, Brown TW. Ultrasonic evaluation of patients with acute right upper quadrant pain. Radiology. 1981; 140(2):449-455.	Observational-Dx	52 patients	Prospective study to define the role of US in acute right upper quadrant pain.	SMS plus stones was most sensitive for AC; 33% had no stones or SMS and were normal.	3
41. Bree RL. Further observations on the usefulness of the sonographic Murphy sign in the evaluation of suspected acute cholecystitis. J Clin Ultrasound. 1995; 23(3):169-172.	Observational-Dx	200 patients	To determine accuracy of SMS in AC.	Sensitivity of SMS 86%, specificity 35%, PPV 43%, NPV 82%. Combination of stones and SMS specificity of 77%. SMS has many false positives.	3

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Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
42. Shea JA, Berlin JA, Escarce JJ, et al. Revised estimates of diagnostic test sensitivity and specificity in suspected biliary tract disease. Arch Intern Med. 1994; 154(22):2573-2581.	Review/Other-Dx	30 articles	To estimate the sensitivity and specificity of diagnostic tests for gallstones and AC.	US has the best unadjusted sensitivity (0.97; 95% CI, 0.95 to 0.99) and specificity (0.95; 95% CI, 0.88 to 1.00) for evaluating patients with suspected gallstones. Adjusted values are 0.84 (0.76 to 0.92) and 0.99 (0.97 to 1.00), respectively. Adjusted and unadjusted results for oral cholecystogram were lower. Radionuclide scanning has the best sensitivity (0.97; 95% CI, 0.96 to 0.98) and specificity (0.90; 95% CI, 0.86 to 0.95) for evaluating patients with suspected AC; test performance is unaffected by delayed imaging. Unadjusted sensitivity and specificity of US in evaluating patients with suspected AC are 0.94 (0.92 to 0.96) and 0.78 (0.61 to 0.96); adjusted values are 0.88 (0.74 to 1.00) and 0.80 (0.62 to 0.98).	4
43. Puc MM, Tran HS, Wry PW, Ross SE. Ultrasound is not a useful screening tool for acute acalculous cholecystitis in critically ill trauma patients. Am Surg. 2002; 68(1):65-69.	Observational-Dx	62 patients	A retrospective study to assess the utility of US in the diagnosis of acute acalculous cholecystitis.	The data revealed a sensitivity of 30% (6/20) and a specificity of 93% (39/42) for US evaluation. 20 patients had subsequent HIDA with a sensitivity of 100% (12/12) and specificity of 88% (7/8). The data do not support US as a reliable routine screening tool for acute acalculous cholecystitis. Despite its convenience as a bedside procedure US has insufficient sensitivity to justify its use and a more sensitive diagnostic tool should be used.	3
44. Shapira-Rootman M, Mahamid A, Reindorp N, Nachtigal A, Zeina AR. Diagnosis of gallbladder perforation by ultrasound. Clin Imaging. 2015;39(5):827-829.	Observational-Dx	11 patients	To identify possible pitfalls in the sonographic evaluation of perforated gallbladders.	Only three patients (27%) were diagnosed correctly with gallbladder perforation by ultrasonography prior to surgery. Gangrenous cholecystitis was reported in 10 cases (90%). Visualization of a wall defect was demonstrated in only five cases (45%) and was associated with a focal echogenic soft-tissue mass.	4

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
45. Richmond BK, DiBaise J, Ziessman H. Utilization of cholecystokinin cholescintigraphy in clinical practice. J Am Coll Surg 2013;217:317-23.	Review/Other-Dx	N/A	To review the controversies surrounding the testing methodology, the method of determining normal vs abnormal CCK-HIDA values, and the data both supporting and questioning its clinical use to select patients for cholecystectomy based on the current available literature. To present evidence-based recommendations for theuse of CCK-HIDA in clinical practice.	N/A	4
46. Fuks D, Mouly C, Robert B, Hajji H, Yzet T, Regimbeau JM. Acute cholecystitis: preoperative CT can help the surgeon consider conversion from laparoscopic to open cholecystectomy. Radiology. 2012; 263(1):128-138.	Observational-Dx	108 patients	To establish whether preoperative CT findings in patients with AC were associated with conversion from laparoscopic to open cholecystectomy in patients with calculous AC.	Conversion occurred in 24 (22%) cases. On preoperative CT images, the absence of gallbladder wall enhancement was associated with the presence of gangrenous AC (sensitivity, 73%). The absence of gallbladder wall enhancement (58% and 40% for conversion and nonconversion, respectively; P=.02) and the presence of a gallstone in the gallbladder infundibulum (78% and 22% for conversion and nonconversion, respectively; P=.04) were associated with AC-related conversion in a multivariate analysis. Interobserver agreement for CT study interpretation was very good (median k value, 0.92; range, 0.76-1.00).	2
47. Hakansson K, Leander P, Ekberg O, Hakansson HO. MR imaging in clinically suspected acute cholecystitis. A comparison with ultrasonography. Acta Radiol. 2000; 41(4):322-328.	Observational-Dx	94 patients	To compare the diagnostic value of fast pulse sequences in MRI with US in patients with clinically suspected AC.	MR diagnoses were AC in 23, gallbladder and common bile duct stones in 3, and other pathologic conditions of the abdomen in 7 and normal in 2 patients. US diagnoses were AC in 17, gallbladder stones in 8, other pathologic conditions of the abdomen in 2, normal in 5 and non-conclusive in 3 patients.	2

Right Upper Quadrant Pain
EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
48. Regan F, Schaefer DC, Smith DP, Petronis JD, Bohlman ME, Magnuson TH. The diagnostic utility of HASTE MRI in the evaluation of acute cholecystitis. Half-Fourier acquisition single-shot turbo SE. J Comput Assist Tomogr. 1998; 22(4):638-642.	Observational-Dx	72 patients	To (a) determine the significance of high signal intensity surrounding the gallbladder as seen on T2-weighted HASTE MRIs in patients with AC and (b) to determine the sensitivity of T2-weighted HASTE MRIs in detecting gallbladder and common bile duct calculi in patients with AC.	Of the 72 patients imaged with HASTE MRI, 55 had cholecystitis based on clinical, sonographic, and/or surgical findings. Of these, 45 had acute and 10 had chronic cholecystitis. HASTE MRI demonstrated MR pericholecystic high signal in 41/45 (91%) of the patients with AC. The sensitivity of HASTE MRI in diagnosing AC was 91%. The specificity was 79%. The PPV was 87%, the NPV was 85%, and the overall accuracy of the test was 89%. Gallbladder stones were seen by HASTE MRI in 38/41 (93%) of patients with acute calculus cholecystitis demonstrated at sonography. Common bile duct stones were demonstrated by HASTE MRI in 7/9 (78%) patients and by sonography in 5/9 (56%) patients with documented choledocholithiasis on conventional cholangiography.	2

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
49. Byott S, Harris I. Rapid acquisition axial and coronal T2 HASTE MR in the evaluation of acute abdominal pain. Eur J Radiol. 85(1):286-290, 2016 Jan.	Observational-Dx	468 cases	To assess T2 HASTE MR in acute abdominal imaging and ascertain if it is a reliable alternative to CT in patients under 60.	468 cases included in the study. 349 were negative for acute abdominal pathology, 116 positive for acute abdominal pathology and 3 were indeterminate. In the MR positive group (n=116), 64 had surgery confirming findings (34 appendicitis, 14 SBO, 3 ovarian torsion, 3 LBO, intussusception, ovarian carcinoma, ovarian dermoid, 2 pelvic inflammatory disease, diverticular abscess, crohns, 4 endoscopy for acute bowel pathology) while 51 were managed conservatively with concordant follow up (4 SBO, 11 diverticulitis, 6 pelvic inflammatory disease, 7 inflammatory bowel disease, 7 colitis, 6 pyelonephritis, 2 cholecystitis, renal abscess, pseudomembranous colitis, splenic haematoma, mesenteric adenitis, 2 pancreatitis, lymphoma, epiploic appendagitis). 1 patient had an MR diagnosis of appendicitis but at laparoscopy a sigmoid diverticular perforation was diagnosed and the appendix was normal. In the MR negative group (n=349), 324 had uneventful follow-up, 22 had negative laparoscopies, while 3 had subsequent appendectomies, with appendicitis on histology (3 days, 10 days and 2 months post scan). In the MR indeterminate group (n=3), one was treated conservatively with uneventful follow up, one had laparoscopic appendectomy with normal appendix on histology, one had laparoscopic appendectomy with acute appendicitis on histology. When MR correlated with clinical follow up (n=468), overall diagnostic accuracy is 99% (463/468). When MR findings correlated with direct visualisation at surgery/endoscopy (n=90), sensitivity is 98% (95% CI) and specificity is 92% (95%	3

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
				CI).	
50. Ahvenjarvi L, Koivukangas V, Jartti A, et al. Diagnostic accuracy of computed tomography imaging of surgically treated acute acalculous cholecystitis in critically ill patients. J Trauma. 2011; 70(1):183-188.	Observational-Dx	127 patients	To determine the usefulness of CT findings in predicting necrotic acute AC in intensive care unit patients.	Abnormal CT findings were present in 96% of all the intensive care unit patients. Higher bile density in the gallbladder body and subserosal edema was associated with an edematous gallbladder (specificity, 93.6%; sensitivity, 23.1%). The most specific findings predicting necrotic acute AC were gas in the gallbladder wall or lumen, lack of gallbladder wall enhancement, and edema around the gallbladder (specificity, 99.2%, 94.9%, and 92.4%, respectively; and sensitivity, 11.1%, 37.5%, and 22.2%, respectively).	3
51. Ziessman HA. Nuclear medicine hepatobiliary imaging. Clin Gastroenterol Hepatol. 2010; 8(2):111-116.	Review/Other-Dx	N/A	To review the most common clinical indications of nuclear medicine hepatobiliary imaging (cholescintigraphy).	HIDA can detect high grade biliary obstruction prior to ductal dilatation; images reveal a persistent hepatogram without biliary clearance due to the high backpressure. HIDA also aids in the diagnosis of partial biliary obstruction due to stones, biliary stricture, and sphincter of Oddi obstruction. It can confirm biliary leakage postcholecystectomy and hepatic transplantation. Calculation of a gallbladder ejection fraction after cholecystokinin infusion is commonly used to diagnose chronic acalculous gallbladder disease.	4

Right Upper Quadrant Pain
EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
52. Chung YH, Choi ER, Kim KM, et al. Can percutaneous cholecystostomy be a definitive management for acute acalculous cholecystitis? J Clin Gastroenterol. 2012; 46(3):216-219.	Observational-Tx	57 patients	To evaluate the safety, efficacy, and long-term outcome of percutaneous cholecystostomy without additional cholecystectomy as a definitive treatment for acute acalculous cholecystitis.	Percutaneous cholecystostomy was technically successful in all patients, and no major complications relating to the procedure were encountered. Symptoms resolved within 4 days in 53/57 (93%) patients. The in-hospital mortality rate was 21% (11/57) and elective cholecystectomy was performed in 18/57 (31%). 28 patients were managed non-operatively and cholecystostomy tubes were subsequently removed. These 28 patients were follow-up over a median 32 months and recurrent cholecystitis occurred in 2 (7%).	2
53. Treinen C, Lomelin D, Krause C, Goede M, Oleynikov D. Acute acalculous cholecystitis in the critically ill: risk factors and surgical strategies. Langenbecks Arch Surg. 2015;400(4):421-427.	Review/Other-Tx	N/A	To examine the three common surgical treatments for AAC (open cholecystectomy (OC), laparoscopic cholecystectomy (LC), or percutaneous cholecystostomy (PC)), their prevalence in current literature, and the perioperative outcomes of these different approaches using a large retrospective database.	LC has proven effective in treating AAC when the risks of general anesthesia and the chance for conversion to OC are low. In critically ill patients with multiple comorbidities, PC or OC may be the only available options. Data in the literature and an examination of outcomes within a national database indicate that for severely ill patients, PC may be safer and met with better outcomes than OC for the healthier set of AAC patients.	4

Right Upper Quadrant Pain
EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
54. Lo LD, Vogelzang RL, Braun MA, Nemcek AA, Jr. Percutaneous cholecystostomy for the diagnosis and treatment of acute calculous and acalculous cholecystitis. J Vasc Interv Radiol. 1995;6(4):629-634.	Observational-Dx	58 patients	To determine the safety, efficacy, and diagnostic utility of percutaneous cholecystostomy in patients with suspected calculous or acalculous cholecystitis.	The gallbladder was successfully catheterized in all 58 patients; 48 patients (83%) had a final diagnosis of acute cholecystitis. Clinical benefit was seen in 26 of 28 patients (93%) with calculous cholecystitis and in 16 of 20 patients (80%) with acalculous disease. The six patients who did not respond had pathologic evidence of transmural inflammation, and five had a gangrenous wall. The gallbladder was excluded as the source of sepsis in 10 patients with suspected acalculous cholecystitis. These patients' conditions did not improve after percutaneous cholecystostomy. Of the 48 patients with cholecystitis, 18 underwent cholecystectomy, 25 recovered and had their catheters removed, and five died of other causes with their catheters in place. There was one major complication, and seven minor complications.	4

Right Upper Quadrant Pain
EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
55. Cherng N, Witkowski ET, Sneider EB, et al. Use of cholecystostomy tubes in the management of patients with primary diagnosis of acute cholecystitis. J Am Coll Surg. 2012; 214(2):196-201.	Observational-Tx	185 patients	To review both surgical cholecystostomy tubes and percutaneous cholecystostomy tubes used to treat patients with AC at a tertiary care center and determine whether there was a benefit to these patients compared with conventional therapy of open conversion or open cholecystectomy.	Mean patient age was 71 years and 80% had =1 comorbidity (mean 2.6). 78% of cholecystostomy tubes were percutaneous cholecystostomy tube placement and 22% were surgical cholecystostomy tube placement. Median length of stay from cholecystostomy tube insertion to discharge was 4 days. The majority (57%) of patients eventually underwent cholecystectomy performed by 20 different surgeons in a median of 63 days post-cholecystostomy tube (range 3 to 1,055 days); of these, 86% underwent laparoscopic cholecystectomy and 13% underwent open conversion or open cholecystectomy. In the radiology and surgical group, 50% and 80% underwent subsequent cholecystectomy, respectively, at a median of 63 and 60 days post-cholecystostomy tube. Whether surgical or percutaneous cholecystostomy tube placement, approximately the same proportion of patients (85% to 86%) underwent laparoscopic cholecystectomy as definitive treatment.	2

Right Upper Quadrant Pain
EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
56. Joseph T, Unver K, Hwang GL, et al. Percutaneous cholecystostomy for acute cholecystitis: ten-year experience. J Vasc Interv Radiol. 2012; 23(1):83-88 e81.	Observational-Tx	106 patients	To review the clinical course of patients with AC treated by percutaneous cholecystostomy, and to identify risk factors retrospectively that predict outcome.	Overall, 72 patients (68%) showed an improvement clinically, whereas 34 (32%) showed no improvement or a clinically worsened condition after cholecystostomy. Patients who presented to the emergency department primarily with AC fared better (84% of patients showed improvement) than inpatients (34% showed improvement; P<.0001). Gallstones were identified in 54% of patients who presented to the emergency department, whereas acalculous cholecystitis was more commonly diagnosed in inpatients (54%). Patients with sepsis had worse outcomes overall (P<.0001). Bacterial bile cultures were analyzed in 95% of patients and showed positive results in 52%, with no overall effect on outcome. There was no correlation between the time of onset of symptoms until antibiotic therapy or cholecystostomy in either group. Long-term outcomes for both groups were better for those who later underwent cholecystectomy (P<.0001).	2

Right Upper Quadrant Pain
EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
57. Melloul E, Denys A, Demartines N, Calmes JM, Schafer M. Percutaneous drainage versus emergency cholecystectomy for the treatment of acute cholecystitis in critically ill patients: does it matter? World J Surg. 2011; 35(4):826-833.	Observational-Tx	42 patients	To compare percutaneous drainage of the gallbladder to emergency cholecystectomy in a well-defined patient group with sepsis related to acute calculous/acalculous cholecystitis.	42 patients [median age = 65.5 years (range = 32-94)] were included; 45% underwent emergency cholecystectomy (10 laparoscopic, 9 open) and 55% percutaneous drainage (n=23). Both patient groups had similar preoperative characteristics. Percutaneous drainage and emergency cholecystectomy were successful in 91% and 100% of patients, respectively. Organ dysfunctions were similarly improved by the third postoperative/postdrainage days. Despite undergoing percutaneous drainage, 2 patients required emergency cholecystectomy due to gangrenous cholecystitis. The conversion rate after laparoscopy was 20%. Overall morbidity was 8.7% after percutaneous drainage and 47% after emergency cholecystectomy (P=0.011). Major morbidity was 0% after percutaneous drainage and 21% after emergency cholecystectomy (P=0.034). The mortality rate was not different (13% after percutaneous drainage and 16% after emergency cholecystectomy, P=1.0) and the deaths were all related to the patients' preexisting disease. Hospital and intensive care unit stays were not different. Recurrent symptoms (17%) occurred only after AC in the percutaneous drainage group.	2

Right Upper Quadrant Pain
EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
58. American College of Radiology. ACR Appropriateness Criteria® Radiation Dose Assessment Introduction. Available at: https://edge.sitecorecloud.io/americancoldf5f-acrorgf92a-productioncb02-3650/media/ACR/Files/Clinical/Appropriateness-Criteria/ACR-Appropriateness-Criteria-Radiation-Dose-Assessment-Introduction.pdf.	Review/Other-Dx	N/A	To provide evidence-based guidelines on exposure of patients to ionizing radiation.	No abstract available.	4

Evidence Table Key

Study Quality Category Definitions

- *Category 1* The study is well-designed and accounts for common biases.
- *Category 2* The study is moderately well-designed and accounts for most common biases.
- *Category 3* There are important study design limitations.
- *Category 4* The study is not useful as primary evidence. The article may not be a clinical study or the study design is invalid, or conclusions are based on expert consensus. For example:
  a. The study does not meet the criteria for or is not a hypothesis-based clinical study (e.g., a book chapter or case report or case series description);
  b. The study may synthesize and draw conclusions about several studies such as a literature review article or book chapter but is not primary evidence;
  c. The study is an expert opinion or consensus document.
- Meta-analysis
  a. *Good quality* – the study design, methods, analysis, and results are valid and the conclusion is supported.
  b. *Inadequate quality* – the study design, analysis, and results lack the methodological rigor to be considered a good meta-analysis study.

Abbreviations Key

Dx = Diagnostic

Tx = Treatment